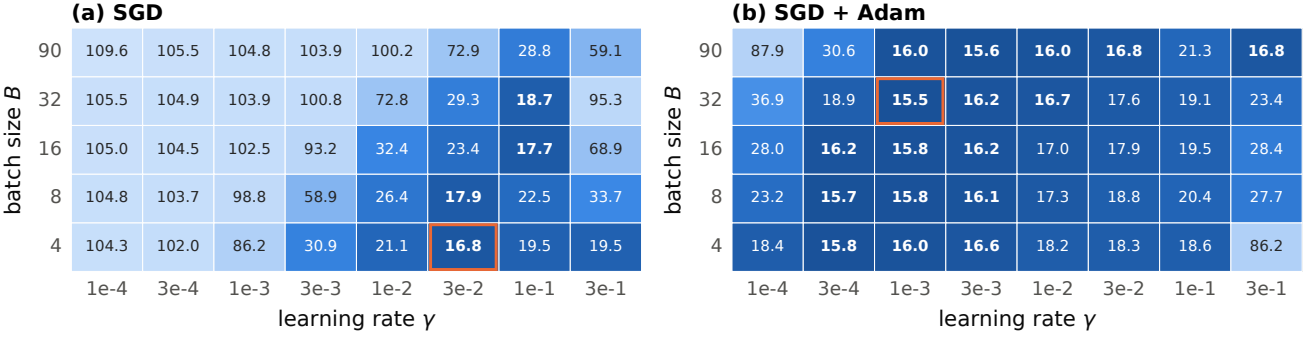
(a) SGD
batch size B
90 109.6 105.5 104.8 103.9 100.2 72.9 28.8 59.1
32 105.5 104.9 103.9 100.8 72.8 29.3 18.7 95.3
16 105.0 104.5 102.5 93.2 32.4 23.4 17.7 68.9
8 104.8 103.7 98.8 58.9 26.4 17.9 22.5 33.7
4 104.3 102.0 86.2 30.9 21.1 16.8 19.5 19.5
1e-4 3e-4 1e-3 3e-3 1e-2 3e-2 1e-1 3e-1
learning rate γ
(b) SGD + Adam
batch size B
90 87.9 30.6 16.0 15.6 16.0 16.8 21.3 16.8
32 36.9 18.9 15.5 16.2 16.7 17.6 19.1 23.4
16 28.0 16.2 15.8 16.2 17.0 17.9 19.5 28.4
8 23.2 15.7 15.8 16.1 17.3 18.8 20.4 27.7
4 18.4 15.8 16.0 16.6 18.2 18.3 18.6 86.2
1e-4 3e-4 1e-3 3e-3 1e-2 3e-2 1e-1 3e-1
learning rate γ